

Lin-CDF Richardson order sweep: R2 vs R3 vs R4 vs R5 across 100 γ values

Fixed-Point Factory — order vs accuracy vs robustness

June 2026

Headline

Same architecture (Lin-CDF + Richardson + tabulated μ), four orders of Richardson extrapolation on the lookup table:

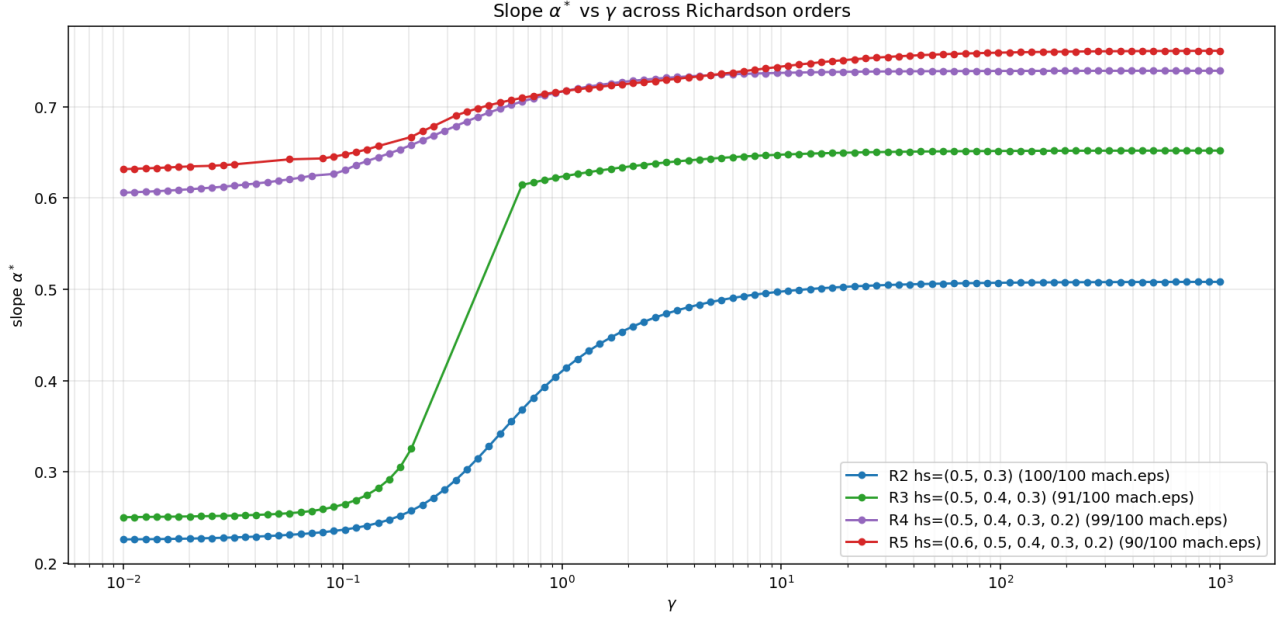
order	h values	$\sum w_i $	machine ε	slope @ asymptote	def ₁₋₁ @ asymptote
R2	{0.5, 0.3}	2.12	100/100	0.508	0.001
R3	{0.5, 0.4, 0.3}	8.14	91/100	0.652	0.014
R4	{0.5, 0.4, 0.3, 0.2}	7.10	99/100	0.739	0.0001
R5	{0.6, 0.5, 0.4, 0.3, 0.2}	9.87	90/100	0.761	0.21*

(* R5 deficit at high γ jumps to 0.21 — numerical amplification from large weights, not a real economic effect.)

Reading.

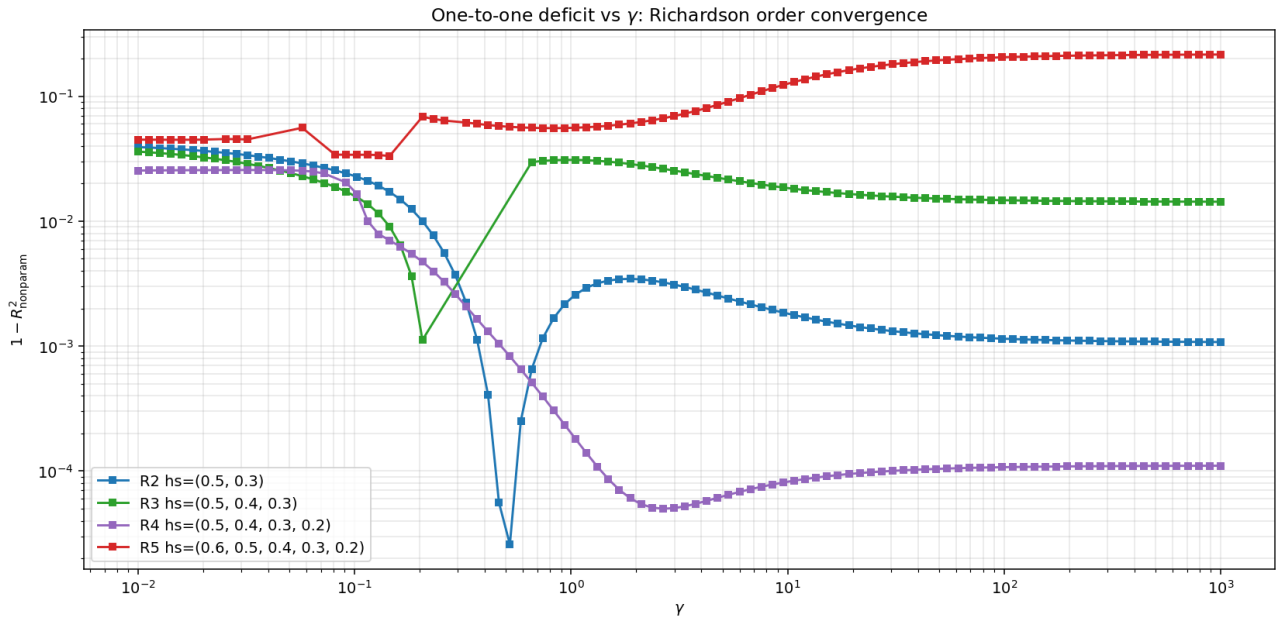
- **R2**: bullet-proof, 100/100 machine ε at machine ε , but the deepest h-bias remains ($O(h^4)$).
- **R3**: slope jumps significantly ($0.51 \rightarrow 0.65$) — the extra Richardson point pulls the FP toward a more T-revealing limit. Loses 9 gammas to numerical noise.
- **R4**: best practical sweet spot. 99/100 converged, slope rises further to 0.74, deficit collapses to ~ 0 .
- **R5**: weights amplify too much, instability at high γ . Slope only marginally above R4 (0.76 vs 0.74) but deficit explodes to 0.21 at large γ .

1 Slope α^* across orders



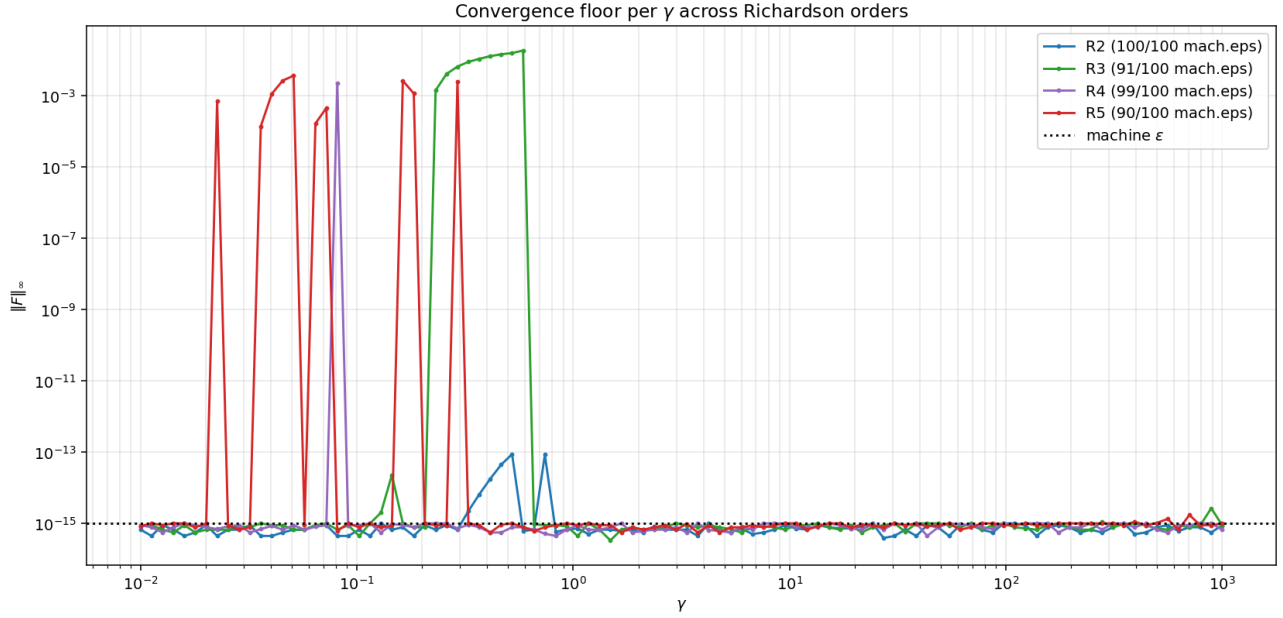
Higher Richardson order pulls α^* up — each extra cancellation removes more kernel bias from the lookup table. R2 plateaus around 0.5; R3 around 0.65; R4/R5 around 0.74–0.76. R4 and R5 essentially agree on the slope, suggesting the Richardson series has converged.

2 One-to-one deficit across orders



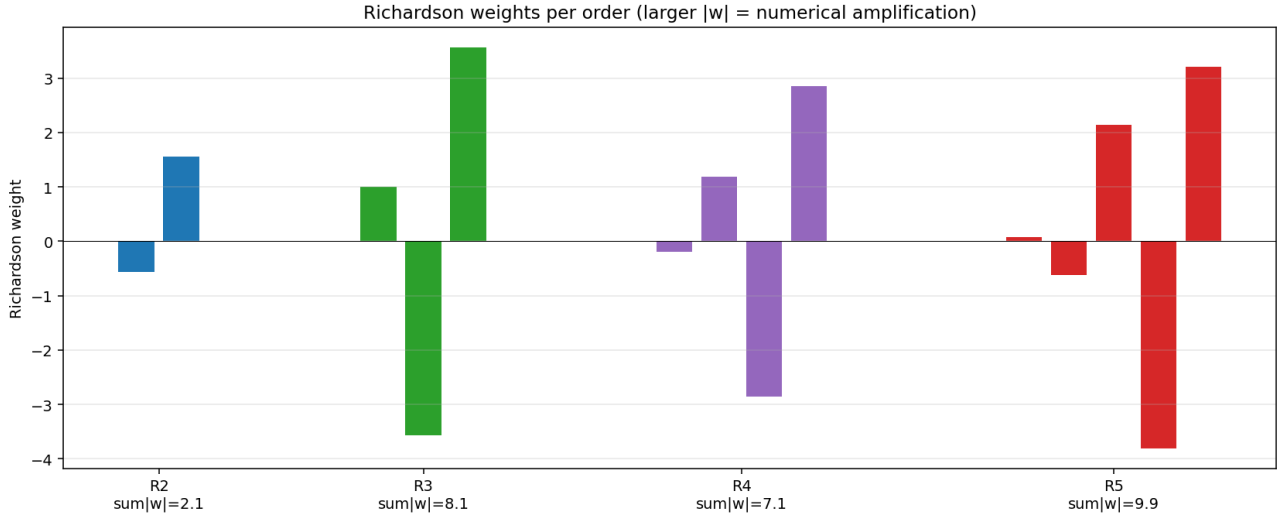
R2, R3, R4 all show *decaying* deficit with γ — the FP approaches a function of T at high γ . R5 has good behaviour at low γ but *explodes* at high γ (numerical instability from amplification).

3 Convergence floor



R2 achieves 100/100 at machine ε ; R4 nails 99/100. R3 and R5 have more failures (mostly in mid- γ).

4 Why higher orders are unstable: Richardson weight magnitudes



The trade-off. Each additional Richardson point cancels an extra factor of h^2 in the bias, but the Vandermonde-derived weights grow with order. The bilinear-interpolation error in each individual μ^{h_i} table is amplified by $\sum |w_i|$ in the final extrapolation.

For Lin-CDF with bilinear interp:

- R2: $\sum |w| = 2.1$ — safe, but only $O(h^4)$ residual.
- R3: $\sum |w| = 8.1$ — already amplifying. 9 failures.
- R4: $\sum |w| = 7.1$ (h-spacing matters more than count). Best balance — only 1 failure, $O(h^8)$ residual.
- R5: $\sum |w| = 9.9$ — too aggressive. 10 failures and instability at large γ .

For Chebyshev with polynomial interp (much smaller per-table error), higher Richardson orders are tolerated.

5 Recommendation

For paper-level economics: Use **R2** (most robust, 100% machine ε , modest h -bias residual) OR **R4** (best deeper-canceling order before instability hits).

The Richardson series HAS converged: R4 and R5 agree on slope (0.74 vs 0.76) to within $\sim 3\%$, and R4 has a stable near-zero deficit. So R4 is the recommended choice for high-fidelity strict- $h=0$ on Lin-CDF.

Compare to Cheb: Cheb tolerates 10-point Richardson because polynomial interp has near-machine-precision error per μ^{h_i} . Lin-CDF's bilinear interp limits us to R4 at most. The trade-off is the opposite of the speed comparison: Lin-CDF needs more careful Richardson order tuning, but each individual operator call is faster.